

Lab 1

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Servers (1)
 - PostgreSQL 18
 - Databases (2)
 - dvdrental**
 - postgres
 - Login/Group Roles
 - Tablespaces

dvdrental/postgres@PostgreSQL 18*

dvdrental/postgres@PostgreSQL 18

Query Query History

```
1 with rankedfilms as(  
2   select c.name as category, f.title as film_title, f.length as film_len,  
3   dense_rank() over (partition by c.name order by f.length desc, f.title) as rn  
4   from film f  
5   join film_category fc on f.film_id = fc.film_id  
6   join category c on c.category_id = fc.category_id  
7 )  
8  
9 select * from rankedfilms where rn <= 3
```

Data Output Messages Notifications

Showing rows: 1 to 48 Page No: 1 of 1

	category character varying (25)	film_title character varying (255)	film_len smallint	m bigint
1	Action	Dam Forrester	185	1
2	Action	Worst Banger	185	2
3	Action	Casualties Encino	179	3
4	Animation	Gangs Pride	185	1
5	Animation	Pond Seattle	185	2
6	Animation	Sons Interview	184	3
7	Children	Fury Murder	178	1
8	Children	Wrong Behavior	178	2
9	Children	Empire Malkovich	177	3
10	Classics	Conspiracy Spirit	184	1

Total rows: 48 Query complete 00:00:00.114 CRLF Ln 9, Col 39

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Data Output Messages Notifications

Showing rows: 1 to 48 Page No: 1 of 1

	category character varying (25)	film_title character varying (255)	film_len smallint	m bigint
11	Classics	Dracula Crystal	176	2
12	Classics	Lights Deer	174	3
13	Comedy	Control Anthem	185	1
14	Comedy	Saturn Name	182	2
15	Comedy	Searchers Wait	182	3
16	Documentary	Wife Turn	183	1
17	Documentary	Young Language	183	2
18	Documentary	Cause Date	179	3
19	Drama	Jacket Frisco	181	1
20	Drama	Something Duck	180	2
21	Drama	Slacker Liaisons	179	3
22	Family	King Evolution	184	1
23	Family	Baked Cleopatra	182	2
24	Family	Manchurian Curtain	177	3
25	Foreign	Crystal Breaking	184	1
26	Foreign	Sorority Queen	184	2
27	Foreign	Catch Amistad	183	3
28	Games	Chicago North	185	1
29	Games	Moonwalker Fool	184	2

Total rows: 48 Query complete 00:00:00.114 CRLF Ln 9, Col 39

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Data Output Messages Notifications

Showing rows: 1 to 48

	category	film_title	film_len	m
	character varying (25)	character varying (255)	smallint	bigint
30	Games	Monsoon Cause	182	3
31	Horror	Analyze Hoosiers	181	1
32	Horror	Love Suicides	181	2
33	Horror	Texas Watch	179	3
34	Music	Home Pity	185	1
35	Music	Scalawag Duck	183	2
36	Music	Reds Pocus	182	3
37	New	Frontier Cabin	183	1
38	New	Runaway Tenenbaums	181	2
39	New	Wild Apollo	181	3
40	Sci-Fi	Soldiers Evolution	185	1
41	Sci-Fi	Express Lonely	178	2
42	Sci-Fi	Connecticut Tramp	172	3
43	Sports	Smoochy Control	184	1
44	Sports	Records Zorro	182	2
45	Sports	Star Operation	181	3
46	Travel	Muscle Bright	185	1
47	Travel	Sweet Brotherhood	185	2
48	Travel	Born Spinal	179	3

Total rows: 48 Query complete 00:00:00.114 CRLF Ln 9, Col 39

Lab 2

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dvdrental/postgres@PostgreSQL 18

Query Query History

```

1  with m as (
2      select DATE_TRUNC('month', payment_date) as month, sum(amount) as revenue
3      from payment
4      group by 1
5  )
6  select
7      month, revenue,
8      lag(revenue) over (order by month) as prev_revenue,
9      round((revenue - lag(revenue) over (order by month))
10         / lag(revenue) over (order by month) * 100, 2) as growth_pct
11  from m
12  order by month;
```

Data Output Messages Notifications

Showing rows: 1 to 4

	month	revenue	prev_revenue	growth_pct
	timestamp without time zone	numeric	numeric	numeric
1	2007-02-01 00:00:00	8351.84	[null]	[null]
2	2007-03-01 00:00:00	23886.56	8351.84	186.00
3	2007-04-01 00:00:00	28559.46	23886.56	19.56
4	2007-05-01 00:00:00	514.18	28559.46	-98.20

Total rows: 4 Query complete 00:00:00.105 CRLF Ln 12, Col 16

Lab 3

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Query

```

1 select customer_id, payment_date, amount,
2       sum(amount) over (
3         partition by customer_id
4         order by payment_date
5       ) as running_clv
6 from payment
7 order by customer_id, payment_date;

```

Data Output

Showing rows: 1 to 1000 of 15

	customer_id smallint	payment_date timestamp without time zone	amount numeric (5,2)	running_clv numeric
1	1	2007-02-14 23:22:38.996577	5.99	5.99
2	1	2007-02-15 16:31:19.996577	0.99	6.98
3	1	2007-02-15 19:37:12.996577	9.99	16.97
4	1	2007-02-16 13:47:23.996577	4.99	21.96
5	1	2007-02-18 07:10:14.996577	4.99	26.95
6	1	2007-02-18 12:02:25.996577	0.99	27.94
7	1	2007-02-21 04:53:11.996577	3.99	31.93
8	1	2007-03-01 07:19:30.996577	4.99	36.92
9	1	2007-03-02 14:05:18.996577	3.99	40.91
10	1	2007-03-02 16:30:04.996577	0.99	41.90
11	1	2007-03-17 11:06:20.006577	4.00	46.90

Total rows: 14596 Query complete 00:00:00.182 CRLF Ln 7, Col 36

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Query

```

1 -- Step 1
2 with CustomerSpend as (
3   select cu.customer_id, cu.first_name, cu.last_name, co.country,
4         sum(p.amount) as total_spent
5   from customer cu
6   join address a on cu.address_id = a.address_id
7   join city ci on a.city_id = ci.city_id
8   join country co on ci.country_id = co.country_id
9   join payment p on cu.customer_id = p.customer_id
10  group by cu.customer_id, cu.first_name, cu.last_name, co.country
11 ),
12 -- Step 2
13 RankedSpend as (
14   select *,
15         rank() over (partition by country order by total_spent desc) as spend_rank
16   from CustomerSpend
17 )
18 -- Step 3
19 select country, first_name, last_name, total_spent, customer_id
20 from RankedSpend
21 where spend_rank = 1
22 order by country;

```

Data Output

Showing rows: 1 to 108 of 1

country character varying (50)	first_name character varying (45)	last_name character varying (45)	total_spent numeric	customer_id integer
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Total rows: 108 Query complete 00:00:00.139 CRLF Ln 22, Col 18

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No limit

Data Output Messages Notifications

SQL Showing rows: 1 to 108 Page No: 1 of 1

	country character varying (50)	first_name character varying (45)	last_name character varying (45)	total_spent numeric	customer_id integer
1	Afghanistan	Vera	Mccoy	67.82	218
2	Algeria	June	Carroll	151.68	176
3	American Samoa	Anthony	Schwab	47.85	320
4	Angola	Claude	Herzog	93.80	528
5	Anguilla	Bobby	Boudreau	99.68	381
6	Argentina	Jordan	Archuleta	129.71	560
7	Armenia	Stephanie	Mitchell	118.75	41
8	Austria	Nora	Herrera	113.74	266
9	Azerbaijan	Raymond	Mcwhorter	108.75	334
10	Bahrain	Seth	Hannon	108.76	590
11	Bangladesh	Michelle	Clark	146.68	21
12	Belarus	Clara	Shaw	189.60	144
13	Bolivia	Joel	Francisco	92.78	431
14	Brazil	Marion	Snyder	194.61	178
15	Brunei	Lois	Butler	107.66	91
16	Bulgaria	Tyrone	Asher	112.76	540
17	Cambodia	Elmer	Noe	92.76	516
18	Cameroon	Albert	Crouse	96.78	352
19	Canada	Curtis	Irby	167.62	410

Total rows: 108 Query complete 00:00:00.139 CRLF Ln 22, Col 18